

Rithesh Kumar

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Summary

Research Engineer specializing in ML inference systems, model optimization, and production AI infrastructure. Authored openvino-go, Go/CGO bindings for Intel OpenVINO Runtime, and built production gRPC serving infrastructure for embeddings and LLMs, along with a grammar-to-protobuf compiler and a REST-to-gRPC service generator. Master's in Computer Science with research in medical imaging and AI explainability. Experience at Goldman Sachs and Accretional translating ML research into high-performance inference systems at scale.

Work Experience

Founding Engineer - Infrastructure & Platform

Aug. 2025 - Present

Accretional

San Francisco, CA

- Authored openvino-go, an open-source Go/CGO library for Intel OpenVINO Runtime with async inference API, variable state management, and benchmarked 40% latency reduction over ONNX Runtime; built production gRPC EmbeddingService on top, powering semantic search and RAG for AI agents
- Engineered hybrid vector and full-text search stack using sqlite-vec, implementing chunked tokenization with L2-normalized mean pooling for embedding quality, improving semantic retrieval accuracy by 40% over keyword-only baselines
- Built vLLM-based LLM serving infrastructure with Go gRPC routing layer and Modal/Docker GPU deployment, supporting named model server management and OpenAI-compatible streaming completions
- Applied model quantization and caching across inference pipelines, reducing serving latency by 50%; owned systems from database internals and ML inference to deployment pipelines end-to-end
- Designed distributed gRPC service mesh with type-safe protobuf registry and dynamic dispatch, enabling AI agents to route requests and retrieve context across a multi-service inference platform

AI Engineering Intern

Jun. 2024 - Aug. 2024

Accretional

San Francisco, CA

- Built RAG pipeline by retrieving documentation and code examples before generation, reducing hallucinations by 35% compared to base model outputs
- Applied LoRA fine-tuning to adapt LLMs for code generation and rebuilt the prompt system in a VS Code extension, enabling users to deploy generated APIs to cloud functions in under 30 seconds

Machine Learning Scientist

Jul. 2020 - Jul. 2023

Goldman Sachs

Bengaluru, India

- Developed LSTM models for loan repayment forecasting and optimized inference pipelines through model quantization and caching, achieving 15% accuracy improvement and reducing latency by 60% from 200ms to 80ms
- Led real-time loan processing pipeline by implementing caching and circuit breakers, handling 1M+ requests per month with 99.9% uptime and graceful degradation under load
- Built A/B testing framework and introduced CI/CD processes across ML team, reducing deployment risk by 80% and cutting deployment time by 70%
- Built simulation models to assess COVID-19 impact on lending risk, enabling data-driven decisions during unprecedented scenarios when historical patterns were unreliable

Machine Learning Research Intern

May 2019 - Jul. 2019

Sprinklr

Gurgaon, India

- Built sentiment analysis pipelines for multilingual, fragmented social media text, achieving 85% accuracy, and applied pruning and quantization to cut model size by 70% while retaining 95% of original accuracy for production deployment

Education

Master of Science in Computer Science

Sep. 2023 - Jun. 2025

University of California, Santa Cruz

Santa Cruz, CA

- GPA: 3.9/4.0. Focused on machine learning and AI, particularly medical imaging applications with emphasis on clinical trust and explainability
- Research in CT scan denoising and AI explainability, developing methods that improve image quality by 30% while maintaining diagnostic accuracy
- Explored attention mechanisms and visualization techniques, creating interpretable models that increase clinician confidence by 60%

- GPA: 9.4/10.0, Graduated with Honors. Published work on automated prostate cancer grading achieving 92.38% accuracy, contributing to Springer proceedings
- Collaborated with pathologists to identify key diagnostic features, designing models that achieved 85% agreement with expert assessments

Projects

Formal Grammars as Protocol Buffers

Jun. 2026

Go, Protocol Buffers, gRPC, EBNF, Compilers

GitHub

- Built a compiler that converts formal grammars into typed protobuf schemas and applied it across 10+ formats, including CSS, HTML, SVG, HTTP, Dockerfile, JSON, YAML, Starlark, and XML, implementing parsing and rendering as reflection over the generated schema
- Represented the ARM64 instruction set as protobuf messages generated from ARM's official specification, covering 634 instruction classes and 4,355 encodings, and verified 228 classes, including the undocumented Apple AMX coprocessor, on real hardware

REST APIs as Typed gRPC Services

May 2026

Go, Protocol Buffers, gRPC, OpenAPI

GitHub

- Built a converter that generates typed protobuf and gRPC service definitions from any OpenAPI v3 specification, with each RPC implemented as a call to the underlying REST endpoint
- Used the converter to wrap the OpenAI (35 gRPC services, 242 RPC methods) and Cloudflare (about 220 services) REST APIs into typed gRPC services that regenerate directly from the upstream spec

OpenVINO Go Bindings

Feb. 2026

Go, C++, CGO, OpenVINO

GitHub

- Built idiomatic Go/CGO bindings for Intel OpenVINO Runtime using a three-layer architecture (Go API, CGO bridge, C++ wrapper), with async inference and variable state management for recurrent models
- Benchmarked 40% latency reduction over synchronous ONNX Runtime; delivered a production-ready open-source library with 90%+ test coverage, now used as a dependency in production embedding and RAG services

gRPC Embedding & LLM Serving Pipeline

2026

Go, gRPC, vLLM, OpenVINO, Python

GitHub

- Built production EmbeddingService (rpcembed) with chunked tokenization, mean pooling, and L2 normalization for long-text embedding quality, backed by openvino-go for hardware-accelerated inference
- Developed vllm-service, a Go gRPC manager for vLLM inference processes with named model server routing, OpenAI-compatible streaming, and Modal/Docker GPU deployment

Publications

Machine Learning, Image Processing, Network Security and Data Sciences

2023

Prostate Cancer Grading Using Multistage Deep Neural Networks

Springer

- Proposed a three-stage ensemble deep learning method for automatic prostate cancer grading, achieving 92.38% classification accuracy on 10,000 histopathological images
- Developed a pipeline using UNet segmentation followed by an ensemble of Xception, ResNet-50, and EfficientNet-b7, outperforming single-model baselines by 8%

Security in Computing and Communications (SSCC 2018)

2019

Network Anomaly Detection Using ANNs Optimised with PSO-DE Hybrid

Springer

- Developed a network anomaly detection system using ANNs optimized with a hybrid PSO-DE algorithm, achieving 94% detection accuracy and cutting false positives by 30% versus signature-based methods
- Improved ANN training convergence by 25% and detection performance by 12% through the combined PSO-DE optimization approach

Skills

Programming Languages Python, Go, C++, CGO, TypeScript, Scala**Machine Learning & AI** OpenVINO, vLLM, LLM Inference, PyTorch, LoRA/PEFT, RAG, Model Optimization, Vector Search**Inference & Systems** Async Inference, Model Quantization, gRPC, Protocol Buffers, EBNF/Compiler Design**Cloud & Infrastructure** AWS, Docker, Modal, Google Cloud, CI/CD, Cloudflare, Envoy**Data & Storage** SQLite, sqlite-vec, PostgreSQL, Redis, Vector Databases, NumPy/Pandas